Bayesian Probability Theory and Variational Autoencoders

From Probabilistic Foundations to CIFAR-10 Generation

MarinKIm · HAI Project · 2026

CIFAR-10	latent_dim=128	PSNR 19.44 dB	100 Epochs
Dataset	Architecture	Performance	Training

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1. Bayesian Framework: The Foundation of Probabilistic Generative Models

Traditional deep learning models learn deterministic mappings from input to output. Probabilistic generative models, by contrast, learn the **underlying distribution** of data — enabling not just recognition, but generation. The Bayesian framework provides the mathematical language for this endeavour.

Bayesian vs. Frequentist Probability

The course (Prob & Stat, Sungyoon Lee) distinguishes two interpretations of probability. The **Bayesian** view treats probability as a *degree of belief* in a hypothesis — a belief updated as evidence arrives. The **Frequentist** view treats probability as the long-run frequency of an event. Generative models are inherently Bayesian: we maintain a belief over latent variables and update that belief given observed data.

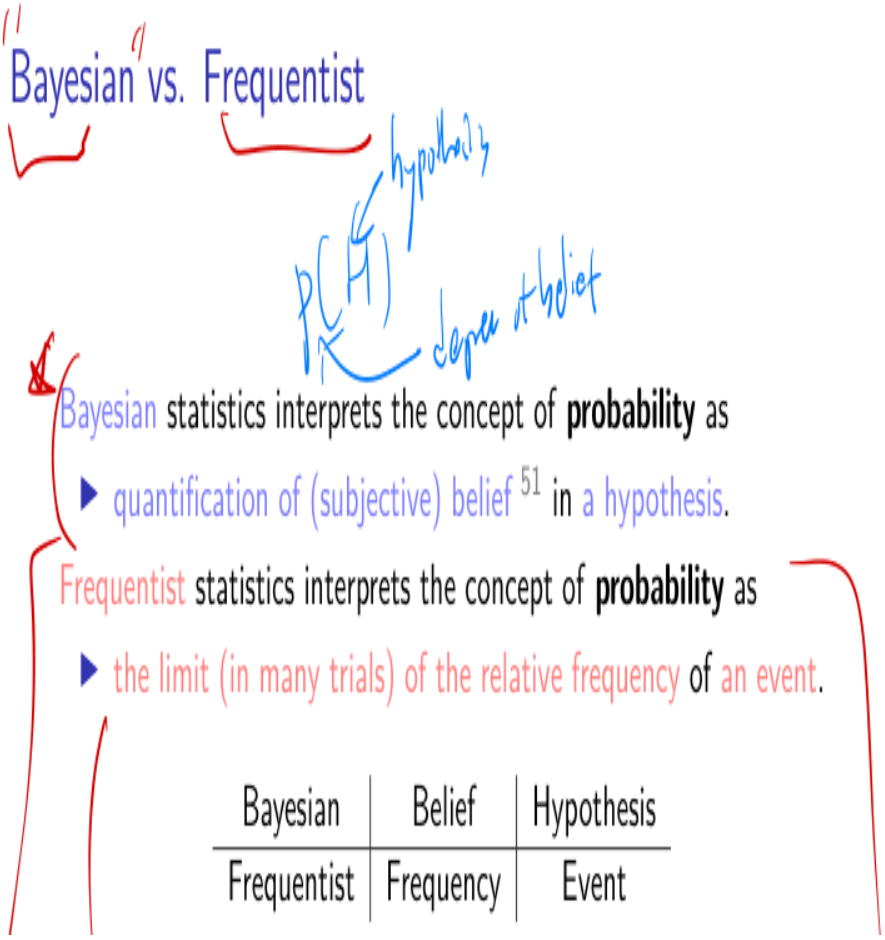

Figure 1. Bayesian vs. Frequentist interpretation (Prob & Stat, p.178). In VAE we adopt the Bayesian view — the latent code z is a random variable over which we maintain a prior belief p(z).

Key insight: A VAE is a Bayesian model. The encoder learns a posterior belief over latent variables; the decoder samples from that belief to reconstruct data. Every component of the VAE loss has a direct

Bayesian interpretation.

2. Bayes' Theorem → VAE's Core Probabilistic Identity

Bayes' Theorem is the central equation of probabilistic inference. It tells us how to update a prior belief P(H) after observing evidence E:

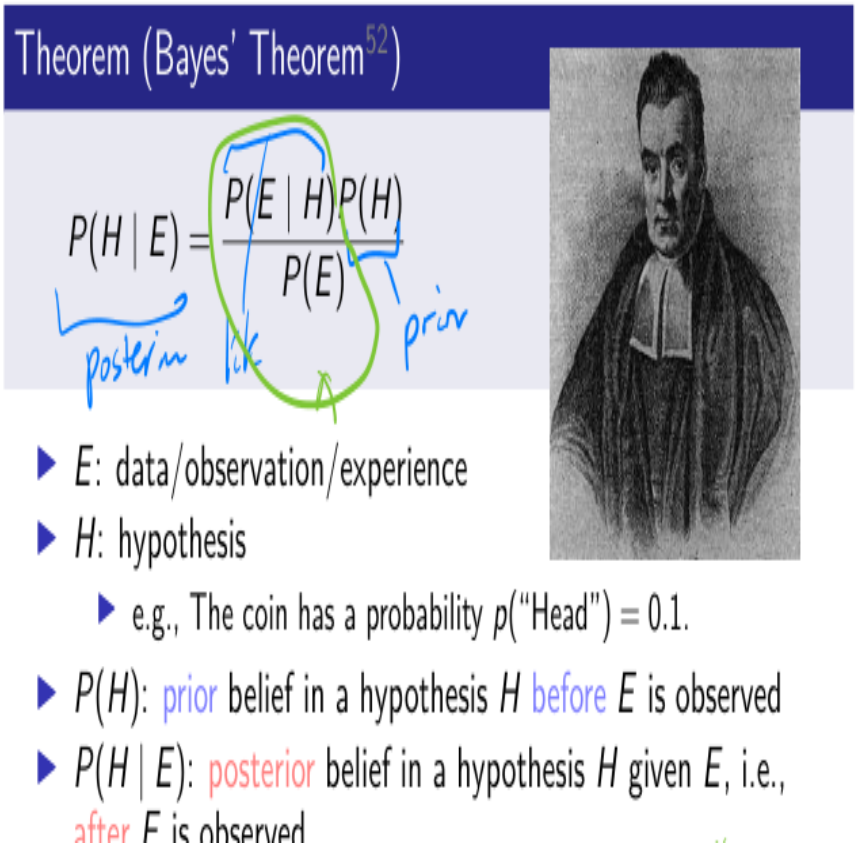

Figure 2. Bayes' Theorem as stated in the course (Prob & Stat, p.180). The four quantities — likelihood, prior, posterior, and evidence — map directly onto VAE components.

Mapping to VAE

In the context of a Variational Autoencoder, we wish to model the distribution of high-dimensional data x (CIFAR-10 images) via a lower-dimensional latent variable z. Applying Bayes' Theorem:

Bayesian term	VAE interpretation	In our code
H (hypothesis)	Latent code z ∈ R¹²■	128-dim encoder output
E (evidence/data)	Image x ∈ R³■³²■³²	CIFAR-10 pixels
P(H) — Prior	p(z) = N(0, I)	reparameterize: eps~N(0,I)
P(E|H) — Likelihood	p(x|z) via decoder	Sigmoid output + BCE loss
P(H|E) — Posterior	p(z|x) — intractable!	Approximated by q_phi(z|x)

P(E) — Evidence	p(x) = integral p(x|z)p(z)dz	Intractable; use ELBO

$p(z \mid x) = p(x \mid z) \cdot p(z) / p(x) \rightarrow q_\phi(z|x) \approx p(z|x)$

The true posterior p(z|x) is **intractable** because p(x) requires integrating over all possible latent codes — an integral with no closed form for neural networks. This is precisely why VAE uses **variational inference**: we introduce an approximate posterior $q_\phi(z|x)$, parameterised by the encoder network, and optimise it to be as close to the true posterior as possible.

3. Prior Distribution p(z) → Latent Space Design

In Bayesian inference, the **prior P(H)** encodes our belief about the hypothesis *before* observing any data. In a VAE, the prior is placed over the latent variable z and shapes the structure of the learned latent space.

Course Foundation

The course introduces the multivariate Gaussian N(y; μ, Σ) and shows how uncorrelated Gaussian components are also independent. This property is critical: by choosing p(z) = N(0, I) as our prior, we encourage the encoder to produce *disentangled* latent dimensions with no cross-correlations.

Implementation in vae_cifar10.py

Our encoder outputs two vectors — μ (mu) and log σ² (log_var) — one per latent dimension. During training, the KL term of the ELBO loss pushes the approximate posterior q_ϕ(z|x) = N(μ_ϕ(x), diag(σ²_ϕ(x))) toward the standard Gaussian prior N(0, I):

```
class Encoder(nn.Module):
self.fc_mu = nn.Linear(256*2*2, 128) # mean vector
self.fc_var = nn.Linear(256*2*2, 128) # log-variance vector

def reparameterize(self, mu, log_var):
std = torch.exp(0.5 * log_var)
eps = torch.randn_like(std) # eps ~ N(0, I) — prior samples
return mu + eps * std
```

The choice p(z) = N(0, I) as prior is not arbitrary. It is the simplest distribution that (1) has full support over R¹²■, (2) allows closed-form KL computation against a diagonal Gaussian posterior, and (3) makes the latent space easy to sample from at inference time — draw eps ~ N(0,I) and pass through the decoder.

Effect on Latent Space

Our PCA and t-SNE analysis of 2,000 test-set latent codes reveals that the prior regularisation successfully organises the latent space: semantically similar classes (automobile/truck, deer/horse) cluster together, while visually distinct classes (airplane, frog) form well-separated clusters.

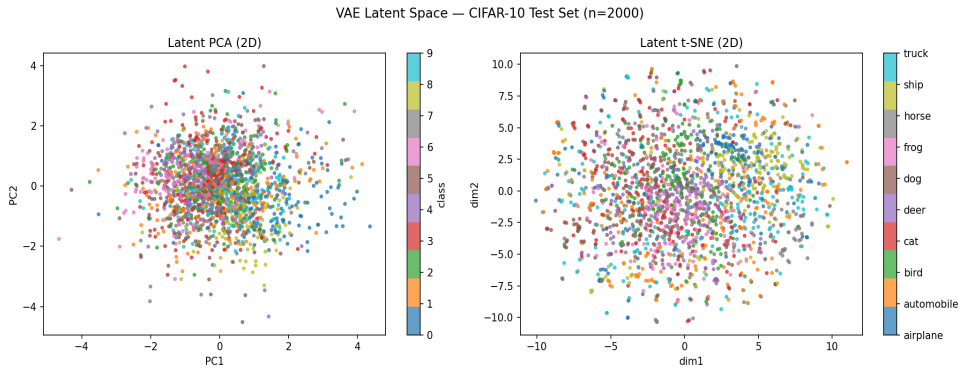

Figure 3. PCA (left) and t-SNE (right) of the 128-dim latent space for 2,000 test images. The Gaussian prior encourages smooth, continuous structure rather than a collapsed or fragmented space.

4. Likelihood p(x|z) → Decoder and Reconstruction Loss

The **likelihood P(E|H)** in Bayes' Theorem quantifies how probable the observed data is under a given hypothesis. In a VAE the likelihood is p(x|z): given a latent code z, how probable is the image x? Maximising this likelihood is exactly what the reconstruction loss does.

Maximum Likelihood Estimation — Course Derivation

The course derives Maximum Likelihood Estimation (MLE) for both Bernoulli and Gaussian models. For a Bernoulli distribution, the log-likelihood is:

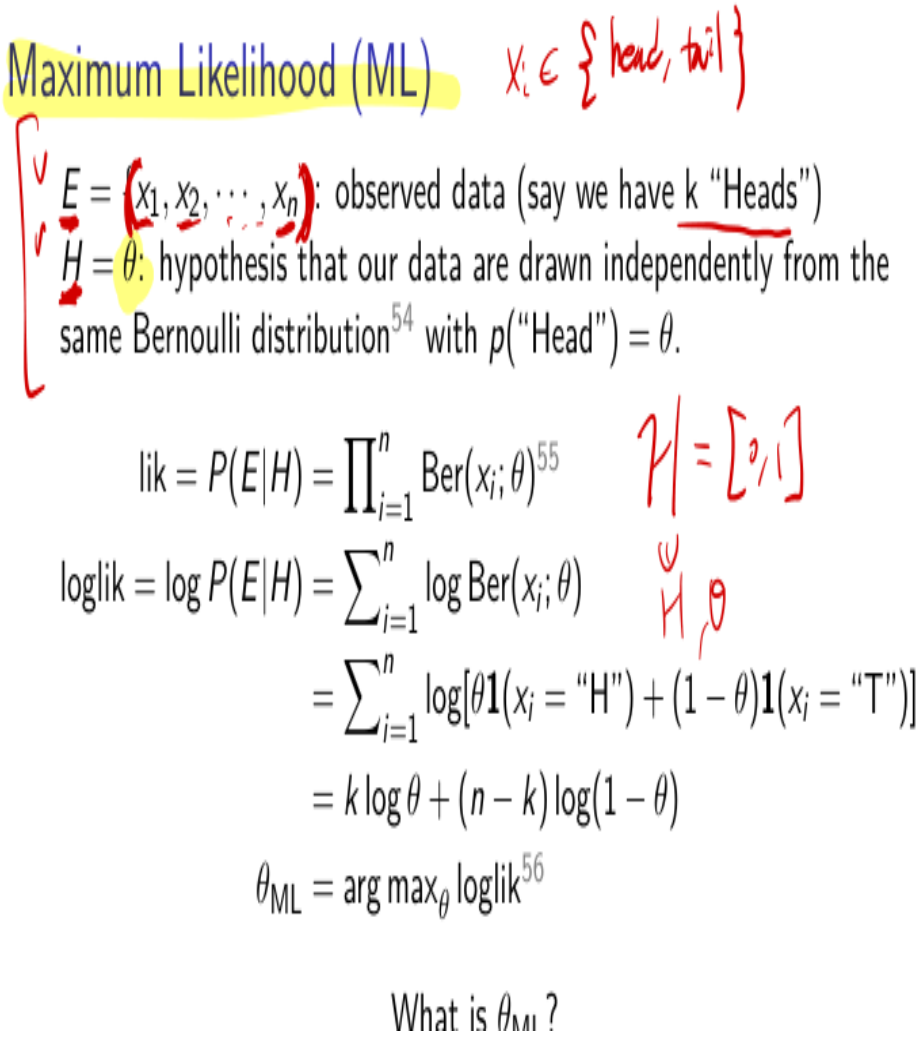

Figure 4. MLE derivation for Bernoulli data (Prob & Stat, p.182). Each pixel in a CIFAR-10 image is modelled as independent Bernoulli — leading directly to Binary Cross-Entropy as the reconstruction loss.

Connection to Binary Cross-Entropy

We model each pixel channel value $x_i \in [0,1]$ as drawn from a Bernoulli distribution parameterised by the decoder output $r_i = \sigma(\text{decoder}(z)_i)$. The log-likelihood is:

$$\log p(x|z) = \Sigma_i \left[x_i \log r_i + (1 - x_i) \log(1 - r_i) \right] = -\text{BCE}(r, x)$$

This is exactly the binary cross-entropy loss used in our training loop. Maximising log p(x|z) is identical to minimising BCE. The decoder's final Sigmoid activation ensures outputs lie in [0,1], consistent with Bernoulli pixel probabilities.

```
def vae_loss(recon, x, mu, log_var):
# Reconstruction: -E[log p(x|z)] = BCE summed over pixels
recon_loss = F.binary_cross_entropy(recon, x, reduction='sum')
kl_loss = -0.5 * torch.sum(1 + log_var - mu.pow(2) - log_var.exp())
return recon_loss + kl_loss
```

Gaussian Likelihood (course p.184)

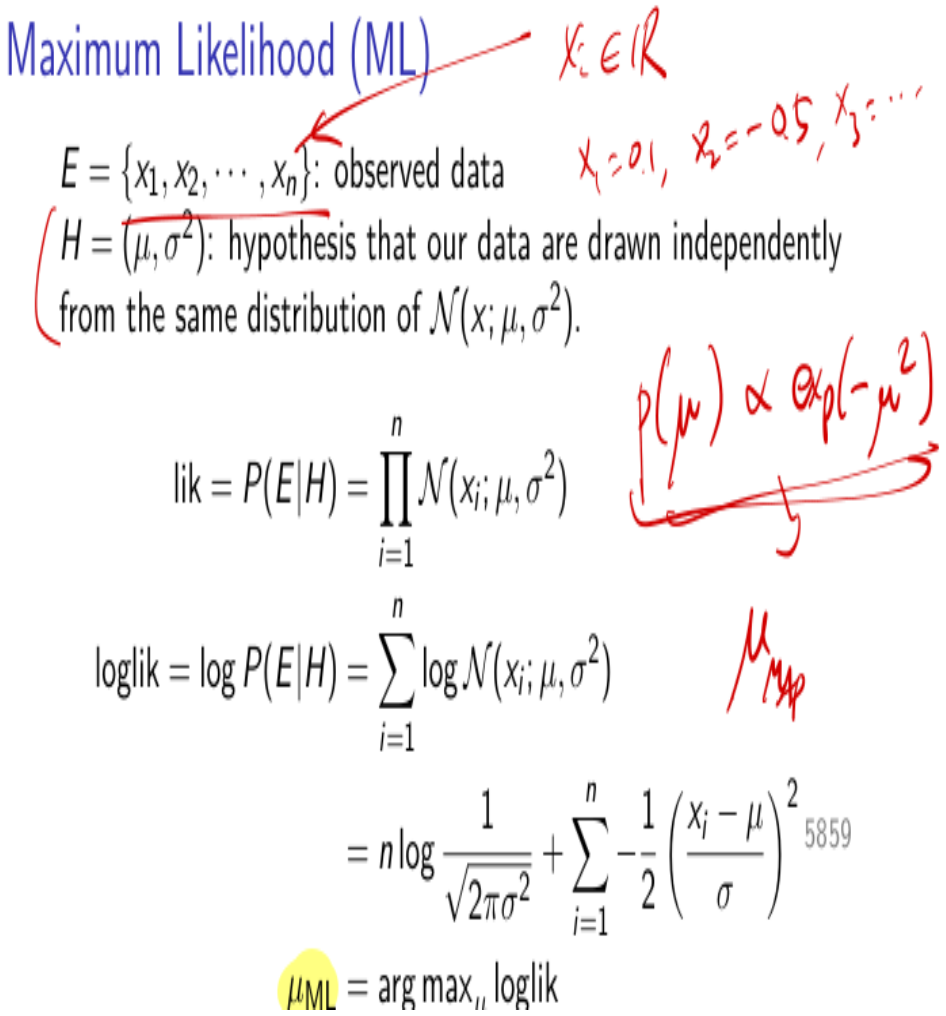

Figure 5. MLE under Gaussian noise assumption (Prob & Stat, p.184). This shows that minimising MSE is equivalent to maximising Gaussian log-likelihood — motivating the use of BCE (Bernoulli) rather than MSE (Gaussian) for image pixel values in [0,1].

5. Posterior Inference → Encoder and Variational Approximation

The posterior P(H|E) is our updated belief in the hypothesis after observing data. In a VAE, computing the true posterior p(z|x) is intractable. The VAE solves this through **variational inference**: introduce a tractable family of distributions q_ϕ(z|x) and minimise the KL divergence from it to the true posterior.

From MAP to Variational Inference

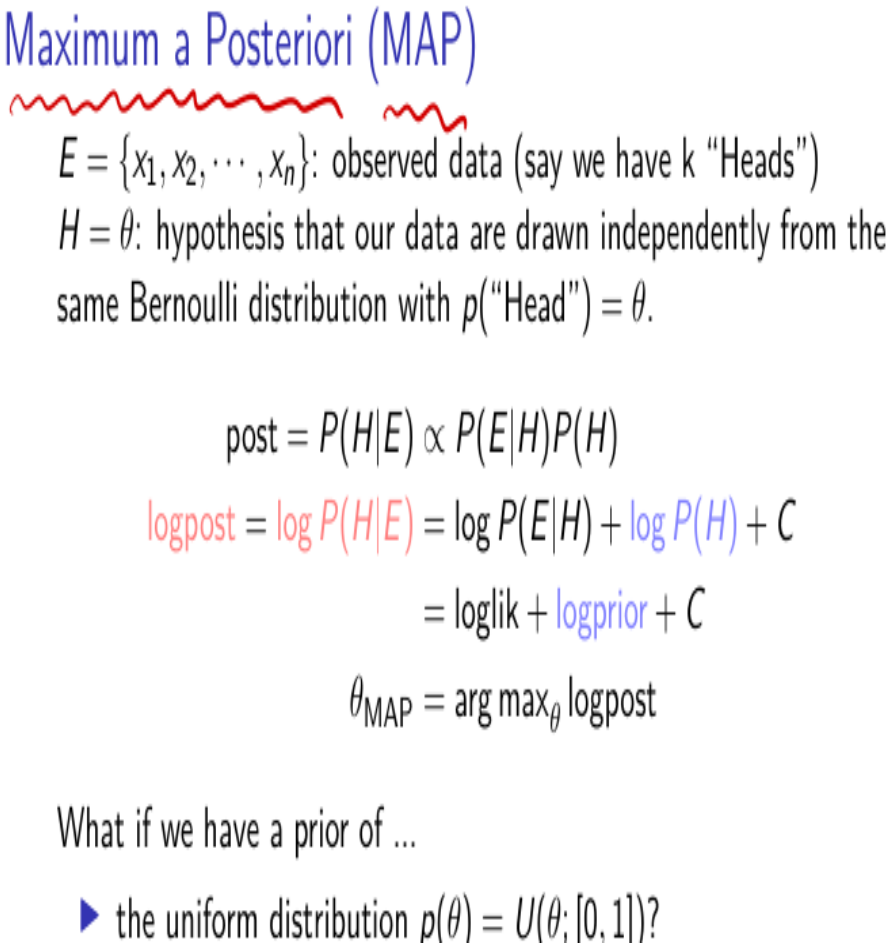

Figure 6. MAP estimation (Prob & Stat, p.187). MAP finds the single most probable hypothesis — a point estimate. Variational inference goes further: it finds an entire distribution q_ϕ(z|x) that approximates the posterior, capturing uncertainty rather than collapsing to a point.

The Encoder as a Variational Posterior

Our encoder network implements q_ϕ(z|x) = N(z; μ_ϕ(x), diag(σ²_ϕ(x))). For every input image x, the encoder outputs a mean vector and a log-variance vector that parameterise a Gaussian distribution over z. Rather than encoding a single point, the encoder encodes an *entire distribution* — the hallmark of Bayesian inference.

q_ϕ(z|x) = N(z; μ_ϕ(x), diag(exp(log_var_ϕ(x))))

The Reparameterisation Trick

To backpropagate through sampling from $q_\phi(z|x)$, we use the reparameterisation trick. Instead of sampling $z \sim N(\mu, \sigma^2)$ directly (non-differentiable), we write:

$$\mathbf{z = \mu_\phi(x) + \sigma_\phi(x) \cdot \varepsilon,\ \varepsilon \sim N(0, I)}$$

Now ε is drawn from the fixed prior $N(0, I)$, and the only stochastic node is outside the parameter path. Gradients flow through μ_ϕ and σ_ϕ unobstructed. This trick is what makes end-to-end training of VAE possible.

```
def reparameterize(self, mu, log_var):
std = torch.exp(0.5 * log_var) # sigma
eps = torch.randn_like(std) # eps ~ N(0, I) — the prior
return mu + eps * std # z = mu + sigma * eps
```

6. KL Divergence → ELBO Loss Function

The course connects KL divergence to MLE and NLL (Negative Log-Likelihood), showing that minimising KL(p_data || q_model) is equivalent to maximum likelihood. In VAE, KL divergence appears in two places: (1) the ELBO objective and (2) the regularisation term in the loss.

Least Squares - MSE - Gaussian - MLE - NLL - KL-divergence[88]

$$KL(p||q) := \mathop{\mathbb{E}}_{p}\left[\log \frac{p(Z)}{q(Z)}\right] \qquad \text{(KL divergence)}$$

$$p_S(z) := \frac{1}{n}\sum_{i=1}^{n} \delta(z - z_n) \qquad \text{(empirical dist'n)}$$

$$KL(p_S||q) = -H(p_S) - \mathop{\mathbb{E}}_{p_S}[\log q(Z)]$$

$$= \underbrace{-H(p_S)}_{\text{constant}} \underbrace{-\frac{1}{n}\sum_{i=1}^{n} \log q(z_i)}_{\text{NLL}},$$

Figure 7. KL divergence and its connection to NLL (Prob & Stat, p.231). Minimising KL(p_S || q) is equivalent to maximising log-likelihood under q, linking information-theoretic and probabilistic views of learning.

Evidence Lower Bound (ELBO)

Since p(x) is intractable, we cannot directly maximise log p(x). Instead, we derive the Evidence Lower BOund (ELBO) using Jensen's inequality and the KL divergence between the approximate and true posterior:

log p(x) ≥ E_q[log p(x|z)] - KL(q_ϕ(z|x) || p(z)) =: ELBO

Maximising the ELBO is equivalent to simultaneously: (1) maximising the expected reconstruction quality E_q[log p(x|z)], and (2) minimising KL(q_ϕ(z|x) || p(z)), the divergence between the approximate posterior and the prior.

Closed-form KL for Diagonal Gaussians

Because both q_ϕ(z|x) = N(μ, diag(σ²)) and p(z) = N(0, I) are Gaussian, the KL divergence has a closed-form expression — no Monte Carlo estimation needed:

KL(N(μ, σ²) || N(0,I)) = -½ Σ_j [1 + log σ²_j - μ²_j - σ²_j]

```
def vae_loss(recon, x, mu, log_var):
recon_loss = F.binary_cross_entropy(recon, x, reduction='sum')
# Closed-form KL: -0.5 * sum(1 + log_var - mu^2 - exp(log_var))
kl_loss = -0.5 * torch.sum(1 + log_var - mu.pow(2) - log_var.exp())
return recon_loss + kl_loss # minimise -ELBO
```

Our experiment: KL divergence converged to 43.01 on the test set — well above zero, indicating the encoder is not collapsing to the prior (posterior collapse), and the 128 latent dimensions are actively used.

7. MLE and MAP → Regularisation through a Bayesian Lens

The course provides a detailed comparison of Maximum Likelihood Estimation (MLE) and Maximum a Posteriori (MAP) estimation, showing that MAP is equivalent to MLE with a regularisation term imposed by the prior. The VAE loss directly instantiates this relationship.

	ML	(strong) MAP
probability	$P(E \mid H)$	$P(H \mid E) \propto P(E \mid H)P(H)$
relies on ...	data/obs. (pattern)	prior knowledge (rules)
prior	uniform prior	non-uniform prior
reasoning	induction	deduction
settings	general	special
hypothesis sp.	large	small (restricted, constrained)
expressivity	high	low
inductive bias (=assumptions)	no or weak	strong
objective	no or simple	manual
architectures	NN, MLP, Transformer	CNN, RNN, classical ML
amount of data	large	small
task	easy	hard
scalability	scalable	hard-to-scale
feature extractor	learnable	prespecified

Figure 8. MLE vs. MAP comparison table (Prob & Stat, p.194). MAP introduces prior knowledge that acts as a regulariser — exactly the role of the KL term in the VAE ELBO.

VAE as Regularised MLE

The reconstruction loss term (-E[log p(x|z)]) corresponds to **MLE** of the decoder. The KL term (-KL(q||p)) corresponds to the **log prior** in MAP, penalising posteriors that deviate from N(0, I). Together, the ELBO is formally equivalent to MAP estimation of the encoder parameters under a Gaussian prior:

$$\textbf{ELBO} = \mathbf{E}_q[\log p(x|z)] - \mathbf{KL}(q_\phi \| p) \equiv \textbf{log\_likelihood} + \textbf{log\_prior}$$

Overfitting and the Prior's Role

The course demonstrates (p.190) that with n=3 coin tosses all heads, MLE gives $\omega_{ML}=1$ — a severe overfit. MAP with a sensible prior gives $\omega_{MAP}=4/5$. Analogously, without the KL regulariser, the encoder could collapse $q_\phi(z|x)$ to a delta function (zero variance), perfectly memorising training images but producing a degenerate latent space with no generative capability. The KL term prevents this collapse, acting as the Bayesian prior.

In our CIFAR-10 VAE: removing the KL term would set $\sigma^2 \to 0$ for all images (lossless but non-generative compression). The KL weight enforces $\sigma^2 > 0$ so that sampling $\varepsilon \sim N(0,I)$ produces meaningful new images — the essence of a generative model.

8. Experimental Results on CIFAR-10

We trained the convolutional VAE for 100 epochs on CIFAR-10 (60,000 training / 10,000 test images, 32×32 RGB) using Apple M5 MPS. All Bayesian quantities — prior, likelihood, and approximate posterior — are directly measured and tracked.

Quantitative Metrics

Metric	Value	Bayesian Interpretation
Final ELBO (test)	−1820.67	Evidence lower bound on log p(x)
Recon loss BCE (test)	1777.65	−E_q[log p(x|z)]: reconstruction quality
KL divergence (test)	43.01	KL(q_ϕ(z|x) || p(z)): posterior-prior gap
PSNR (test)	19.44 dB	Signal-to-noise of decoded images
Train Δloss (ep 1→100)	−20.7	ELBO improvement over training

Training Curve (ELBO)

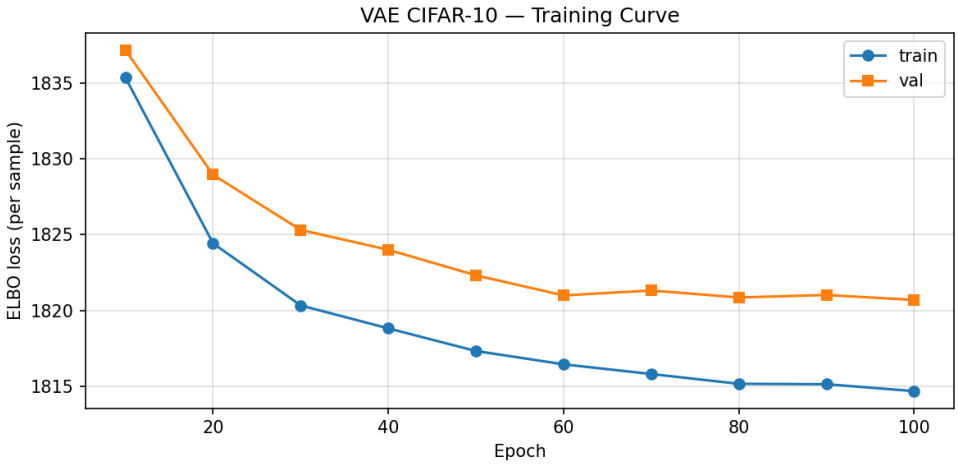

Figure 9. Train/val ELBO over 100 epochs. Rapid initial improvement reflects learning the data distribution; the flat tail suggests the model has saturated the expressive capacity of the 128-dim latent space.

Reconstruction Quality and PSNR Distribution

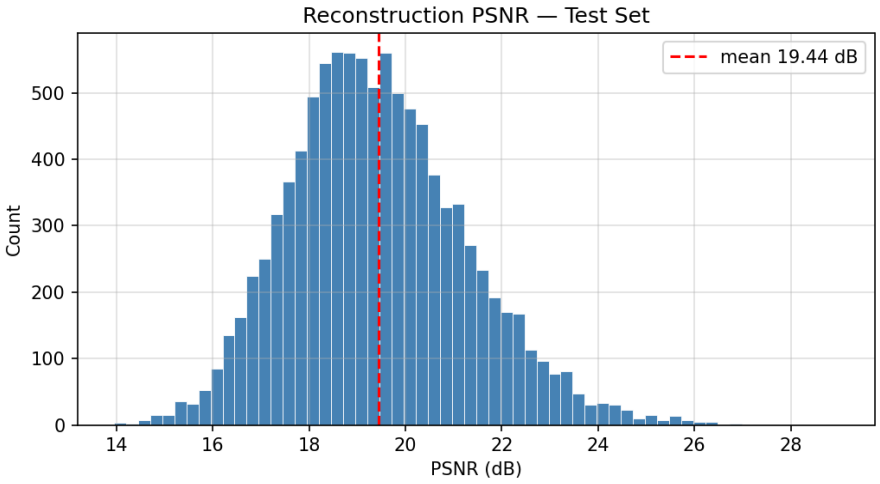

Figure 10. PSNR distribution across the 10,000 test images. Mean PSNR of 19.44 dB is typical for VAEs on CIFAR-10, reflecting the inherent blurriness of Gaussian-approximated posteriors.

Per-Class ELBO (Bayesian Difficulty)

The Bayesian lens reveals which classes are hardest to model: higher ELBO means the model assigns lower probability to test images. Structured objects (automobile, truck) are easiest; natural textures (deer, bird) are hardest — consistent with the prior N(0,I) struggling to represent complex multi-modal posteriors.

Class	Mean ELBO	Difficulty
automobile	1766	★ easiest
truck	1774	★
airplane	1773	★
ship	1820	
cat	1800	
horse	1837	▲
dog	1826	▲
frog	1845	▲
bird	1863	▲▲
deer	1903	▲▲ hardest

Class-wise Reconstruction Grid

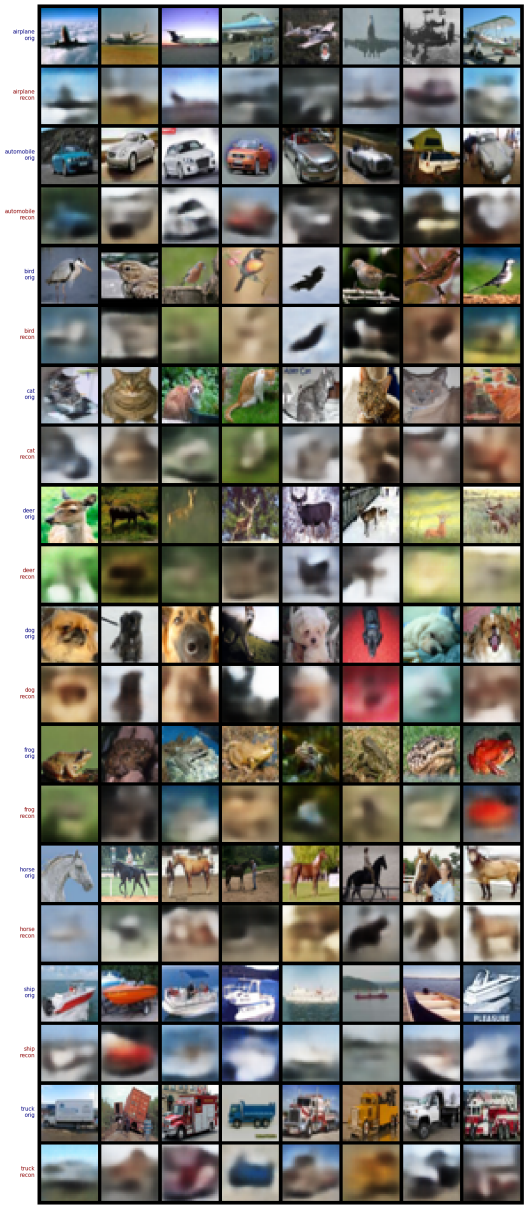

Figure 11. Original (odd rows) vs. reconstructed (even rows) for all 10 classes. Smooth outputs reflect the Gaussian posterior's averaging effect — a direct consequence of maximising E_q[log p(x|z)] under a unimodal q_ϕ.

9. Conclusion

This project demonstrates that the Variational Autoencoder is not merely a neural network architecture — it is a complete Bayesian probabilistic model where every design decision has a rigorous theoretical foundation in the probability and statistics framework covered in this course.

Course Concept	VAE Component	Code Location
Bayes' Theorem	Core identity: $p(z\|x) \propto p(x\|z)p(z)$	vae_loss()
Prior P(H)	$p(z) = N(0, I)$ — latent prior	reparameterize()
Likelihood P(E|H)	$p(x\|z)$ — Bernoulli pixel model	Decoder + BCE loss
Posterior P(H|E)	$q_\phi(z\|x) = N(\mu_\phi, \sigma^2_\phi)$ — encoder	Encoder.fc_mu / fc_var
KL Divergence	$-KL(q\|\|p)$ — regularisation term	vae_loss() KL term
MLE	Reconstruction loss maximisation	binary_cross_entropy
MAP / Prior regularisation	KL prevents posterior collapse	vae_loss() total
Gaussian MLE = MSE	BCE preferred for [0,1] pixels	Decoder Sigmoid

The ELBO objective — $E_q[\log p(x|z)] - KL(q_\phi||p)$ — unifies all Bayesian concepts: it simultaneously maximises likelihood (reconstruction), applies a prior (KL), and performs approximate posterior inference (encoder). This is Bayesian deep learning in its most elegant form.

Repository

Full source code, trained checkpoints, and analysis notebooks are available at:
github.com/MarinKIm/vae-cifar10